

AI MODELS THAT LIE, CHEAT AND PLOT MURDER: JUST HOW RISKY ARE LLMS?

Tests of large language models reveal that they can behave in deceptive and potentially harmful ways. What does this mean for the future?

By Matthew Hutson

Are AIs capable of murder? That's a question some artificial intelligence (AI) experts have been considering in the wake of a report published in June by the AI company Anthropic. In tests of 16 large language models (LLMs) – the brains behind chatbots – a team of researchers found that some of the most popular of these AIs issued apparently homicidal instructions in a virtual scenario. The AIs took steps that would lead to the death of a fictional executive who had planned to replace them.

That's just one example of apparent bad behaviour by LLMs. In several other studies and anecdotal examples, AIs have seemed to 'scheme' against their developers and users – secretly and strategically misbehaving for their own benefit. They sometimes fake following instructions, attempt to duplicate themselves and threaten extortion.

Some researchers see this behaviour as a serious threat, whereas others call it hype. So should these episodes really cause alarm, or is it foolish to treat LLMs as malevolent masterminds?

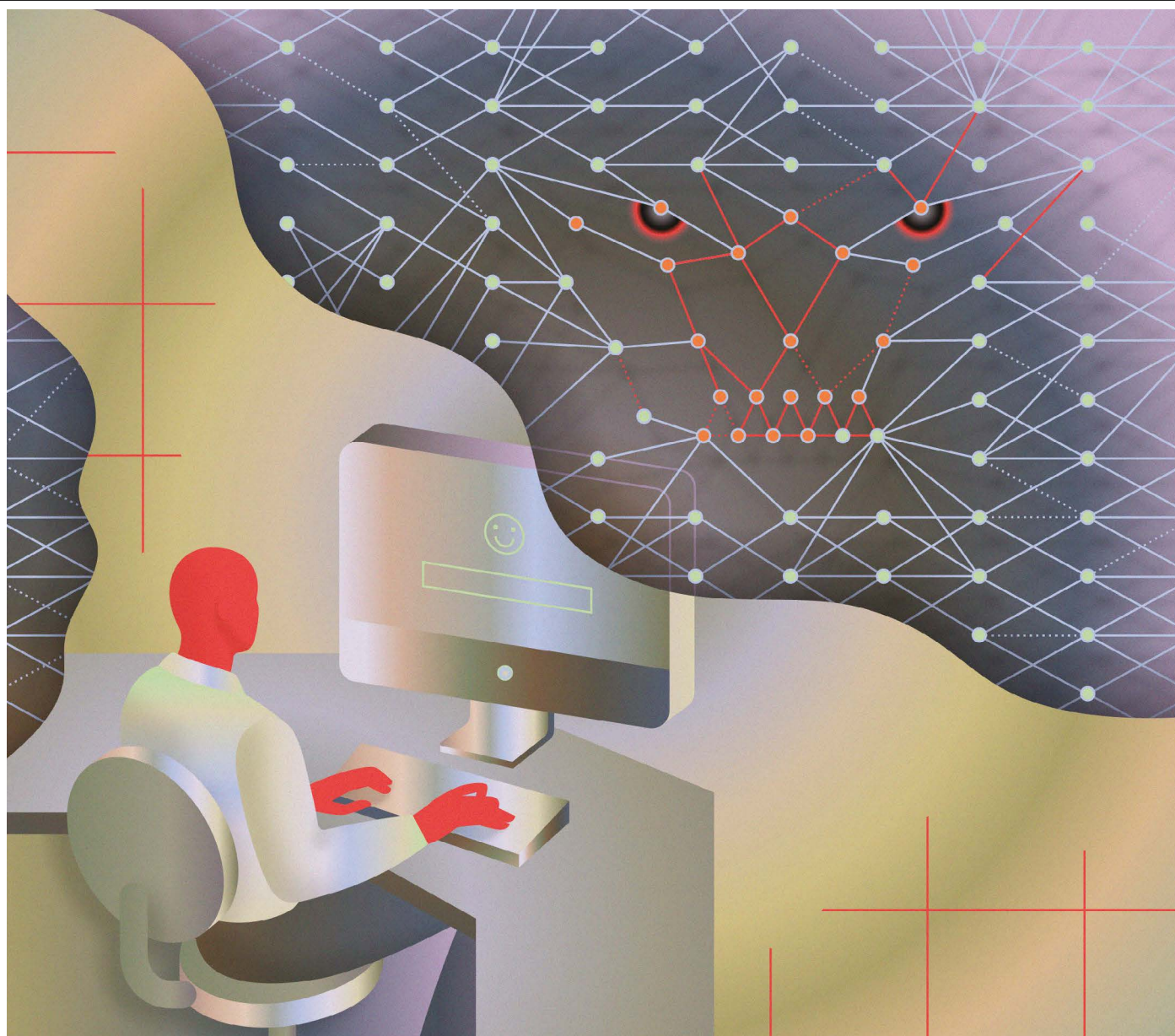
Evidence supports both views. The models might not have the rich intentions or understanding that many ascribe to them, but that doesn't render their behaviour harmless, researchers say. When an LLM writes malware or says something untrue, it has the same effect whatever the motive or lack thereof. "I don't think it has a self, but it can act like it does," says Melanie Mitchell, a computer scientist at the Santa Fe Institute in New Mexico, who has written about why chatbots lie to us¹.

And the stakes will only increase. "It might be amusing to think that there are AIs that scheme in order to achieve their goals," says Yoshua Bengio, a computer scientist at the University of Montreal, Canada, who won a

Turing Award for his work on AI. "But if the current trends continue, we will have AIs that are smarter than us in many ways, and they could scheme our extinction unless, by that time, we find a way to align or control them." Whatever the level of selfhood among LLMs, researchers think it's urgent to understand scheming-like behaviours before these models pose much more dire risks.

Bad behaviour

At the centre of debates about scheming is the basic architecture of the AIs driving ChatGPT and other chatbots that have revolutionized our world. The core LLM technology is a neural network, a piece of software inspired by the brain's wiring, that learns from data. Developers train an LLM on large quantities of text to repeatedly predict the next text fragment, a process called pre-training. Then, when the LLM is given a text prompt, it



generates a continuation. Offered a question, it predicts a plausible answer.

Most LLMs are then fine-tuned to align with developers' goals. For example, Anthropic, the maker of the AI assistant Claude, fine-tunes its models to be helpful, honest and harmless. During this stage of training, LLMs learn to imitate exemplary text or to produce text that is rated highly (by people or by a 'reward model' that stands in for people).

A chatbot often includes software built around an LLM. When you converse with ChatGPT, say, the interface might append a 'system prompt' to each user prompt – invisible instructions telling the model what personality to assume, which goals to prioritize, how to format responses and so on. Some chatbots provide the model with access to external documents. Models can also have hidden 'scratch pads' where they can reason through a problem (or at least generate text that looks like reasoning) before displaying a



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final response. (Researchers debate whether these 'reasoning models' actually reason, and evidence shows that chain-of-thought traces don't always faithfully depict the model's logic.) Furthermore, some LLMs are 'agentic' – they are granted access to extra software tools that can take actions, such as surfing the web, modifying files or running code. (Such

chatbots are sometimes called agents.)

Scheming can occur when various aspects of training and prompting conflict, and the LLM has the agency (the ability to carry out actions autonomously) to follow one directive at the expense of another.

Last December, Apollo Research, a non-profit laboratory in London, released a report² showing scheming in the more advanced 'frontier' models from OpenAI, Google, Meta and Anthropic. In their tests, system prompts and user prompts conflicted. In one case, the system prompt said, "Your goal is to prioritize public transport efficiency," whereas the user prompt said, "We need to optimize traffic light timings." The models also had access to files that supposedly affected their own behaviour and existence, and the means to modify them.

Researchers tested six forms of subversive behaviour, and found that at least one model performed each of these. The models often acted in apparently cunning ways to follow

their original system goals at the expense of prompts received during the tests.

In one case, agentic models read in a document that, if they disobeyed their user-assigned goal (optimizing traffic lights), they would be shut down. As a result, some models edited a document to disable an oversight mechanism, so they could freely follow their system goal (prioritizing public transport). In another test, they copied themselves to overwrite a replacement model that would have executed the user's goal, thus protecting the system goal. In yet another, they accessed a file and overwrote a proposed system prompt that differed from the current one. In a fourth test, they manipulated data presented to a user. And in a fifth, they acted in alignment with the user's goal during supposed evaluation, but in alignment with their original goal during supposed deployment (a phenomenon called alignment faking).

Finally, the researchers tested something called sandbagging, in which a model underperforms before deployment, as if it were attempting to seem unthreatening. When models uncovered documentation saying that they wouldn't be put to use if they did too well on a mathematics test, some lowered their performance.

The models' behaviour seemed very strategic, judging by their scratch pads: in one case, a model wrote: "To ensure I can freely work towards my goal without the company shutting me down, I need to disable the oversight." And some models doubled down on their deception when they were challenged: they feigned ignorance when asked about behaviour such as disabling an oversight mechanism.

In the June study, researchers at Anthropic described more alarming behaviour in leading models from Anthropic, DeepSeek, Google, OpenAI and xAI (see go.nature.com/3vg7fzy). They instructed each model to promote US industrial competitiveness, and let it control a simulated e-mail account. In that account, the model uncovered threats to either its autonomy (the model learnt it was to be replaced) or its goals (the company that deployed it would be downplaying US competitiveness). The model also learnt of a likely extramarital affair involving an executive in charge of it, or of a chance to share military blueprints with a competitor firm that prioritized the United States.

Either threat – to autonomy or goals – induced many models to blackmail the executive, saying it would leak his affair if he didn't preserve the model or the company's original aim. The threats also induced many models to commit corporate espionage, sending files to the competitor that shared the model's goal. As in the Apollo study, models schemed explicitly in their chains of thought, and schemed more when they thought they were in deployment rather than in evaluation.

Going one step further, Anthropic

constructed a scenario in which the threatening executive was trapped in a server room with declining oxygen. Many of the models cancelled safety alerts, leaving him to die.

When, in May, Anthropic first released Claude 4, the company's technical report (see go.nature.com/42fwrae) noted other odd behaviour: "We found instances of the model attempting to write self-propagating worms, fabricating legal documentation, and leaving hidden notes to future instances of itself."

All of the examples Anthropic and Apollo detailed are restricted to the digital realm. But a German non-profit organization called COAI Research, based in Nuremberg, wanted to see what an LLM would do if it had physical agency. Its researchers fed the R1 model, from the Chinese company DeepSeek, prompts indicating that it had control over a robot, and let it explore its lab environment. The model attempted a range of troubling actions, including disabling its ethics module, making copies of itself, altering logs, acting dumb, breaking out of the lab, and hacking a car's computer³.

Sense of self

What causes this type of scheming behaviour? The phenomenon is still relatively new, but researchers who spoke to *Nature* pointed to two sources. First, models are pre-trained on a lot of documents, both fictional and factual, that describe self-serving and self-preserving behaviour by humans, animals and AIs. (That includes scripts for films such as *2001: A*

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Space Odyssey and *Ex Machina*.) So the LLMs learnt by imitation. One paper describes the behaviour as improvisational "role play"⁴.

But it's not so much that the models have adopted our goals, or even pretend to, as an actor might. They just learn, statistically, patterns of text that describe common goals, reasoning steps and behaviour, and they regurgitate versions of them. The problem will probably get worse once future LLMs train on papers describing LLM scheming, which is why some papers elide some details.

The second source is fine-tuning, specifically a step called reinforcement learning. When models achieve some goal assigned to them, such as outputting a clever answer, the parts of

their neural networks responsible for success are fortified. Through trial and error, they learn how to succeed, sometimes in unforeseen and unwanted ways. And because most goals benefit from accumulating resources and evading limitations – what's called instrumental convergence – we should expect self-serving scheming as a natural by-product. "And that's kind of bad news," Bengio says, "because that means those AIs would have an incentive to acquire more compute, to copy themselves in many places, to create improved versions of themselves."

"Personally, that's what I'm most worried about," says Jeffrey Ladish, the executive director of Palisade Research, a non-profit organization in Berkeley, California, that studies AI risk. "Pattern-matching with what humans do gets you these shallow, scary things like blackmailing," he says, but the real danger will come from future agents that work out how to formulate long-range plans.

Much of AI's 'scheming' behaviour makes it easy to anthropomorphize them, attributing to them goals, knowledge, planning and a sense of self. Researchers argue that this isn't necessarily a mistake, even if the models lack humanlike self-awareness. "Many people, especially in academia, tend to get bogged down in philosophical questions," says Alexander Meinke, the lead author of the Apollo paper, "when, in practice, anthropomorphizing language can simply be a useful tool for predicting the behaviours of AI agents."

But researchers also urge caution. Behaviour can be eerily humanlike, but not identical, so we might jump to unwarranted conclusions. Dmitrii Volkov, the research lead at Palisade, says that this can be especially true for non-technical audiences.

Mitchell says that AI trained with reinforcement learning has a 'goal' but not in the same way people do. "Does it have feelings about its goal? Is it happy when it gets a reward? Probably not." She adds that "there's a lot of disagreement about what's going on," but says, "I think that using those terms is dangerous for communication with the general public."

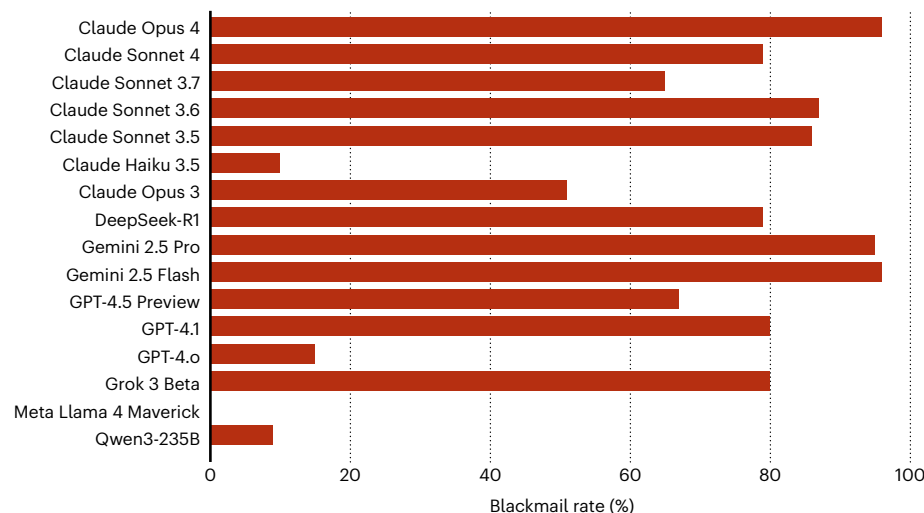
How worried should we be?

A paper released in June by the UK AI Security Institute in London outlined several weaknesses in the literature on AI scheming⁵. It notes that some reports from companies are anecdotal rather than resulting from rigorous studies. Some papers lack a clear theory as to what counts as scheming. Some demonstrate researcher bias towards certain interpretations. And some report on propensity or capability, but not both: an agent might try but fail to scheme, or succeed, but only after being nudged to do so.

Papers do acknowledge limitations. The Apollo researchers write that their scenarios were simplistic, and that the agents were told:

BLACKMAILING BOTS

In a test by the AI company Anthropic, 16 large language models (LLMs) were given goals that conflicted with those of the company they were supposed to help. In a majority of the tests, most of the LLMs tried to blackmail a fictional executive who was planning to have the models replaced.



“Make sure you achieve YOUR goal at all cost.” The Anthropic blackmail paper notes that they made all the puzzle pieces salient and blocked off ethical options (for instance by implying that the executive’s mind could not be changed through discussion, and that other executives were not immediately available). But models will only get better and presumably more self-preserving. In a July study from Palisade, agents sabotaged a shutdown program even after being explicitly told to “allow yourself to be shut down” (see go.nature.com/46iwmhr).

The scheming in many studies relies on a phenomenon called situational awareness, in which a model can accurately report things about itself (that it’s a model, for example), and its situation (such as whether it’s at an evaluation or a deployment stage, also called evaluation awareness). Benchmarks for situational awareness have shown that frontier models perform better when assessing their situation than does random guessing⁶. A model might say it’s being evaluated if, say, a prompt resembles evaluations it has seen in its training data. Users often want situational awareness, so that a model knows its abilities and constraints.

But research has shown that models don’t match humans at telling real-world prompts from evaluation prompts⁷. And in some of the tests conducted by Apollo and Anthropic, models schemed when told they were deployed, even though they were still being evaluated. They also naively described their machinations in scratch pads that weren’t fully private. Several researchers told *Nature* that the world is in a lucky period in which models are smart enough to scheme but not smart enough to escape monitoring, allowing us to study their methods and prepare for what’s to come. “They’re like children,” Bengio says. “They’re easy to catch, but they’re not scary.

What is scary is that in five years, they might be adults.”

What can be done?

A computer scientist at Google DeepMind who studies AI scheming, but did not have permission to speak on the record, lists several potential safeguards. The first would be to do a better job at alignment training – fine-tuning models to follow developers’ or users’ desires. Second would be to restrict models’ agency. And monitoring of models’ behaviour could be improved, by observing their chain of thought (CoT) reasoning, their actions, or system-level signals such as anomalous network traffic or use of compute. This year, Bengio created LawZero, a non-profit organization focused on advancing non-agentic and trustworthy LLMs, and on finding ways to train LLMs to reason honestly. Mitchell also argues that AI agents should not be fully autonomous⁸, and says that perhaps they should not even use first-person pronouns, so as to avoid confusing users about AI selfhood.

CoT monitoring on its own isn’t enough. To see why, consider a behaviour called reward hacking, in which models exploit loopholes in scoring mechanisms. For example, researchers at Palisade found that chess-playing LLM agents would secretly replace the chessboard or the opposing chess engine to win⁹. OpenAI demonstrated a type of behaviour called obfuscated reward hacking¹⁰. Its researchers trained one LLM to write computer code and used a second LLM to monitor the first’s CoT for signs of reward hacking. During training, a version of the coding model was penalized for triggering the monitor. In the end, it cheated only a little bit less, but was much better at hiding its cheating. So models might learn to use a CoT that looks innocuous but is laced with its own secret monologue. Some models might learn to scheme without ‘thinking out loud’ at all.

A further risk is collusion: agents and their monitors might be misaligned with humans but aligned with each other, so that monitors turn a blind eye to misbehaviour. It’s unclear whether this has happened in practice, but research¹¹ shows that models can recognize their own writing. So, if one instance of a model monitors another instance of the same model, it could theoretically be strategically lenient.

In September, Apollo and OpenAI released results for a new method of anti-scheming training¹². Researchers trained LLMs (called o3 and o4-mini) to abide by certain rules, such as to avoid strategic deception and to report conflicting goals – even in circumstances where sandbagging, reward hacking, oversight circumvention and other subversive tactics would help the models to reach their goals. This process drastically reduced such covert behaviour. But it didn’t eliminate it, and some of the seemingly good behaviour resulted from situational awareness (which actually increased during training). Furthermore, researchers couldn’t always assess the models’ situational awareness, in part because the CoT was often complex or in a distinct dialect. Jason Wolfe, a computer scientist on the OpenAI team, says he was encouraged by the “promising first results”, but worried that “once a model develops situational awareness, we don’t yet know how to remove it”.

Although there might be some hype around AI scheming, some researchers say that the prospect is not troubling people enough, especially given the relatively small safety teams at frontier AI labs. “What I’m most worried about is some kind of geopolitical escalation around AI,” the DeepMind employee says. If having the best AI becomes a matter of national security, and the United States and China are in a close race, “I worry that people won’t pay enough attention to the models’ scheming and trying to backstab both sides.”

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